1. TITLE OF THE IDEA

# Apex PdM: Township-Scale Asset Health & Air-Conditioning Predictive Maintenance Platform

A low-cost sensor pod that feels, hears and measures machines, so township teams fix small problems before they become big breakdowns

`LIVE WORKING RIG: https://washer-pdm.pages.dev` **Stage:** Working prototype (TRL 4–5) with live data online

| 11 | 3 | 20×+ | ~RM 170 |
|---|---|---|---|
| WASH SESSIONS RECORDED | SENSORS WORKING IN SYNC | VIBRATION JUMP, IDLE → SPIN | PILOT KIT COST, OFF THE SHELF |

## 2. PROBLEM STATEMENT

**What is the issue?** Air-conditioners, chillers, water pumps and big appliances are usually fixed only **after they break**. Before that, a worn bearing, damper or seal slowly makes the machine shake more, sound different and **work harder, using more electricity**. Nobody notices until the machine stops. By then, a **small part worth tens of ringgit** has turned into a **new motor or compressor worth thousands**, plus days without cooling.

**Who is affected?**

- **Township Communal & Commercial Spaces:** Community **mosques, suraus, kuils (temples), clubhouses, and retail shopping malls** operating many commercial AC compressors daily. Sudden breakdown leaves congregations and shoppers in unbearable tropical heat.
- **Township Facility Management (FM) Teams:** Technicians check machines one by one, by hand, and still cannot see wear happening inside until something fails.
- **Township Residents:** Households facing surprise repair bills, noisy machines and higher electricity bills.

**Why is it important to solve this problem now?** Electricity keeps getting more expensive, and townships are aiming for lower emissions and less waste. A machine that is wearing out uses more power and gets thrown away sooner. Catching the problem early means **a cheap repair instead of a full replacement**, lower bills and less scrap going to landfill.

## 3. PROPOSED SOLUTION

**How does your solution address the problem?** Apex PdM is a small sensor pod that sits on the outside of a machine. It records **vibration, sound and power together**, learns what *normal* looks like for that machine, and gives an **anomaly score and a simple verdict for every run** (normal / check it). A full health score and remaining-life estimate are our next step, once we have recorded real faults.

**1. Vibration (MPU-6050 @ 200 Hz):** 3-axis sensor that feels shaking, unbalance and loose or worn parts.

**2. Sound (INMP441 @ 16 kHz):** Digital microphone that hears grinding, squealing and rattling.

**3. Power (Tapo P110 @ 1 Hz):** Smart plug that measures how hard the motor is working, in watts.

**4. Machine State (planned):** SmartThings reports wash, rinse, spin; saving it with the data and comparing within the same phase is next.

**What technology / product is involved?**

- **Dual-Core ESP32:** One core reads the sensors in real time, the other handles Wi-Fi and cloud upload, so neither slows the other down. Every 5 seconds the pod turns raw signals into **32 signal features**.
- **Keeps Logging Through Wi-Fi Drops:** The pod stores about 4 minutes of data and sends it once the connection is back.
- **Live Web Dashboard:** Anyone can watch the rig live today. A township map and phone alerts are the next build.

## 4. TARGET MARKET / BENEFICIARIES

**Specific Industry or Sector:** Commercial Facility Management (FM), Communal Places of Worship, Retail Real Estate, and Smart Sustainable Townships (e.g., Sime Darby Property's City of Elmina, Bandar Bukit Raja).

- **Communal Places of Worship:** Mosques, Suraus, and Kuils (temples) relying on central air-conditioning split units.
- **Commercial Centers & Clubhouses:** Community gyms, function halls, and retail malls (e.g., Elmina Lakeside Mall).
- **Local Maintenance Contractors (MSMEs):** Technicians who get told which machine needs a look, instead of rushing to emergency calls.

**Urban/Rural Focus:** High-density master-planned urban townships and commercial property clusters.

## 5. UNIQUENESS AND INNOVATION

- **Low Cost:** Industrial vibration sensors cost thousands of ringgit each, so only big factories use them. We spent **about RM 131 in parts for the sensor node** (about RM 170 for a full pilot kit with smart plug) — cheap enough for every AC unit in a mosque or mall.
- **Built to Reduce False Alarms:** One sensor alone gets fooled — a normal fast spin looks like a fault. We check vibration, sound and power **together**, and plan to check against what the machine is doing, before raising an alert.
- **Sees Energy Waste Too:** Because we measure power directly, the system can compare each run's power use against that machine's own normal and show when it starts working harder.
- **Easy to Fit:** Mounts on the outside and plugs in — no rewiring and nothing opened up.

## 6. PROTOTYPE FEASIBILITY, EMPIRICAL EVIDENCE & THE EXPANSION PLAN

**Why we tested on a washing machine:** A front-load washer is one of the toughest machines to monitor. It shakes hard, the load moves around, water sloshes, and the speed keeps changing from slow tumbling to a fast spin. If our pod gets clean, clear data here, a steady-running AC compressor or pump is an easier job. **What we measured** in our 66-minute cycle on 19 Oct: vibration jumped **over 20×** from idle to spin; the main vibration frequency climbed smoothly to **19.9 Hz (about 1,200 rpm)** as the drum sped up; and the washer's sound in the higher bands was **36–90× louder** than the room's background noise, even with a compressor and water pump running nearby.

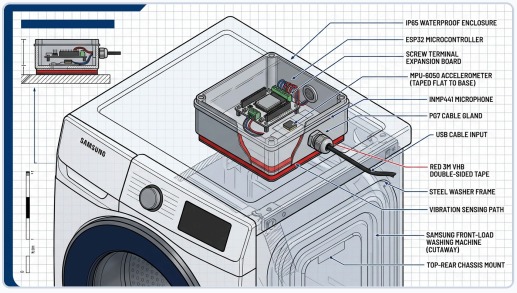



**Figure 1: Sensor pod layout**
• **Hardware:** ESP32 dual-core + MPU-6050 (200 Hz) + INMP441 mic (16 kHz) + Tapo P110 smart plug (1 Hz).
• **Data in Hand: 11 wash sessions recorded** and **8,379 rows** of training-ready data, 32 signal features per 5-second window (firmware-07 window record).
• **Model Validation:** Leave-one-cycle-out test on 8 healthy cycles (earlier snapshot-format data, before the 32-feature windows): 0 falsely flagged (95% upper bound 31%); a simulated doubling of sound was caught in 7 of 8.
• **Live Dashboard:** open to anyone at **https://washer-pdm.pages.dev**.

**From washer to township:** Fans, pumps and compressors wear out the same way — worn parts cause more shaking, then more noise, then more power use, then a breakdown. The same pod and software can move onto **AC units and pumps in township mosques, kuils, community halls and malls**. **Next steps:** record 20+ more cycles to measure our false-alarm rate, then pilot on real township equipment.

## 7. EXPECTED IMPACT, SCALABILITY & COMMERCIAL ROI

**Example (our estimate, to be confirmed in the pilot):**
• One 5 kW commercial AC running 10 hours a day uses about 18,250 kWh a year. If a worn part makes it use just **10% more**, that is ~1,825 kWh — **about RM 900 a year** at ~RM 0.50/kWh.
• A pilot kit costs about **RM 170**, so catching one problem like this **pays for the pod within a few months**.
• Catching a failing part early can also **avoid replacing a whole compressor**, which costs thousands of ringgit.

**Township Quality of Life & ESG Impact:**

- **Less Disruption:** Repairs can be planned ahead, so cooling stays on during Friday prayers, festivals and busy weekends.
- **Less Waste & Lower Emissions:** Fixing one small part instead of replacing a whole unit keeps machines running longer, cuts scrap metal and saves electricity.